

Safeguarding Traditional Knowledge

THE global fight against bio-piracy and preservation of intellectual property rights of the people took a major step forward recently when about 35 countries came together at an international conference to explore options of replicating India's TKDL or Traditional Knowledge Digital Library to protect their traditional knowledge. The *International Conference on Utilization of the Traditional Knowledge Digital Library (TKDL) as a Model for Protection of Traditional Knowledge* held in New Delhi from 22-24 March 2011 demonstrated in ample measure TKDL's immense value in safeguarding national TK.

Acknowledging the success of India's TKDL, Shri P.K. Bansal, Minister of S&T and Earth Sciences, said while inaugurating the conference: "TKDL has made waves around the world, particularly in TK-rich countries, by demonstrating the advantages of proactive action and the power of strong deterrence."

Prof. S.K. Brahmachari, Director General-CSIR said that in the future TKDL could be utilized in open innovation models for coming out with low-cost medicines against diseases such as TB, malaria and filaria. He gave the example of Risorine – an anti-TB medicine currently in the market where bio-enhancer lead was obtained from *Trikatu*. He emphasized that TKDL is community knowledge and that CSIR/AYUSH do not "own" it.

Attended by delegates from about 35 countries including Bangladesh, Bhutan, Botswana, Brazil, Bulgaria, Chile, Cambodia, Indonesia, Iran, Iraq, Jamaica, Kenya, Malaysia, Maldives, Mali, Mongolia, Nepal, Republic of Korea, Russia, Sri Lanka, Thailand, Turkey, Uganda, Vietnam and Zambia, the Conference was aimed at sharing experiences and information on the role of TKDL in documentation of TK, to identify the intellectual property issues in and technical implications of the establishment of a TKDL, and to explore the role and functioning of the TKDL within the international IP protection system. It was organized by the Council of Scientific and Industrial Research (CSIR) in association with

the World Intellectual Property Organization (WIPO).

The TKDL documents traditional knowledge available in public domain in the form of existing literature related to Ayurveda, Unani, Siddha and Yoga, and presents it in searchable and digitized format in English, German, French, Japanese and Spanish. The information is structured to assist patent examiners in their search for existing evidence of previously known/documented knowledge or *prior art*.

Every Indian remembers the feeling of national outrage and personal helplessness as international patents were granted for the wound-healing properties of *haldi* (turmeric) and the anti-fungal properties of neem. The use of *haldi* and neem for such purposes, as Indians protested, was household knowledge. The medicine chests in Indian homes always had a bouquet of medicinal herbs for immediate use. *Haldi* and neem, along with *tulsi* and some other medicinal plants were greatly valued for their healing and health-giving properties.

India did not take this misappropriation lying down. The CSIR decided to challenge the patent—and won. Meanwhile, it became clear that patent examiners were handicapped because often the TK was oral in nature and even when actually documented in manuscripts and treatises (as Ayurveda, Yoga and Siddha were) the information is not structured as per their requirements. It is also not retrievable by search engines and so remains out of the reach of patent examiners, although it is in public domain.

It was against this backdrop that the TKDL was conceived and established. More than 150 experts in traditional medicine, law and computer science spent almost ten years accessing and classifying information codified in authoritative textbooks on Ayurvedic, Unani and Siddha medicines and documenting postures in Yoga. They collated scattered data published in ancient languages and transformed it into a systematic, proprietary database. The TKDL digitized available

information in five languages and in a format that allowed patent examiners to retrieve the information in the way they needed.

As it stands today, the TKDL has documented 85,500 Ayurvedic formulations, 1,20,200 Unani formulations, 13,470 Siddha formulations and 1098 Yoga postures. TKDL safeguards 0.226 million medicinal formulations as these are used in Ayurveda, Unani and Siddha. The TKDL database size runs to 34 million A4 size pages.

The USP of TKDL is that it operates within the established rules and laws of the recognized patent system and does so in a manner that cannot be challenged in any way. TKDL establishes *prior art* and does so in an internationally acceptable manner despite the fact that the source of the *prior art* goes back thousands of years and was originally written in a language few today use regularly. By empowering patent examiners, TKDL is instrumental in preventing the grant of wrong patents.

For instance, a patent that earlier took at least a decade of enormously expensive intensive litigation to be revoked is now being routinely refused in a matter of weeks at no cost (since the patent application is rejected and no rights have been granted that need to be revoked). A new and welcome trend is that many applicants are unilaterally withdrawing their applications on being faced with TKDL—their actions tacit acknowledgement of a misappropriation attempt.

India's leadership in fighting misappropriation of TK began to attract visits of high-level delegations from international agencies. South Africa's Department of Science & Technology visited in 2003 and in 2004, delegations from African Regional Industrial Property Organization visited India with the idea to replicate TKDL for ARIPO Member States. Even five years before the present conference, the Governments of Thailand, Mongolia, Cambodia, Malaysia and Indonesia had already evinced keen interest in TKDL.



On the dais from left :
Shri V.K. Gupta,
Director - TKDL,
Dr. S.K. Brahmachari,
DG - CSIR, and Hon'ble
Minister Shri P.K.
Bansal

concerned. He affirmed that there are solutions beyond TKDL for TK and that every country would have to explore the solution that suits it best. The conference was a way of sharing the Indian TKDL experience; countries were welcome to look closely, go back and use the experience but were under no obligation to replicate the Indian TKDL in toto.

Patent Offices across the world were also eager to have access to TKDL so as to be able to grant patents correctly and without subsequent controversy and expensive and time-consuming litigation. The first request received for TKDL Access was in July 2005. However, formal access to the TKDL was granted with non-disclosure agreement to the European Patent Office only in February 2009 because legal formalities had to be completed.

Once the framework was in place, access was granted to the Indian Patent Office in July 2009; German Patent Office in October 2009; United States Patent and Trademark Office in November 2009, United Kingdom Patent and Trademark Office in February 2010, Canadian Intellectual Property Office in September 2010, and IP Australia in January 2011.

The present conference was scheduled to be the backdrop for signing a similar agreement with the Japan Patent office. However, this was postponed in view of the national calamity in Japan. A similar arrangement is currently being negotiated with the New Zealand IP Office. Interestingly, USPTO patent examiners were trained by CSIR for Search and Examination of TK-based patent applications in October 2010 at the TKDL facility in Ghaziabad.

At the conference, representatives of countries such as Peru, Brazil, Ecuador,

Indonesia, Kenya and Thailand presented brief accounts of the state of TK and/or TK documentation in their countries. Senior officials from WIPO also shared their experiences and detailed the issues related to the grant of a patent. It was clear that TKDL or similar digital database of documentation is a pre-requisite to identify evidence of *prior art*. It was also clear that certain common problems were still to be addressed and that the countries were concerned about these. For example:

- Indian TKDL relied on already documented knowledge or knowledge that was already present in published format. What would be the case with orally transmitted TK?

- Indian documents used for TKDL are considered to belong to all the citizens of the nation. What could be the case for TK held by different communities, even cross-border communities?

- How could secret TK be documented? This was an important question as certain forms of TK are supposed to remain a secret to be effective. Documenting such knowledge would therefore contravene the tenets of TK

- How could the spiritual covenant that travels with the use of TK be preserved?

Wrapping up the session, Shri V.K. Gupta, Chief Architect of the TKDL project, was categorical that there cannot be a "one size fits all" situation as far as TKDL was

However, irrespective of what path a country chose to safeguard its TK, acquiring prior informed consent was a non-negotiable and mandatory requirement. Shri Gupta emphasized that the ownership lies with the TK holder. This is an ethical issue that is beyond question. He drew on his vast experience and advised that the way forward was to take up several initiatives and if even one or two yielded continuous results that get noticed, then, those are the leads to follow. He reiterated that implementing TKDL has worked for India and the next step forward was to utilize and add value to TKDL.

In his closing remarks, Mr Naresh Prasad, Executive Director, WIPO said that the attempt would be to ensure enduring institutional relationship with CSIR subject to demands by member states who have seen TKDL at work. He requested the delegates to mull over all they had heard and seen, reflect and consult and then to give their feedback to WIPO.

The Conference proved that the TKDL story is not just one of significance from the context of national pride but is one of inspirational leadership that can serve as a beacon illuminating the way forward in the global fight against bio-piracy and preservation of IP rights of the people.

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